

[RedCap Unit 1 Report]

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[1. Technical Background]

- [Unit Name] [RedCap FR1 init-time validation]
- [Module Scope] `openair1/PHY/INIT/nr_parms.c`
- [Goal] 在 [PHY init] 階段先驗證 [RedCap FR1] 的基本能力限制，避免非法配置流入後續 [allocator]、[transport path]、[rfsimulator] 執行流程。
- [TS 38.306 Clause 4.2.21.1] 定義 [FR1 最大頻寬 20 MHz]、[1Rx mandatory / 2Rx optional]、[DL layers <= 2]、[UL MIMO not supported]。
- [TS 38.331 Clause 5.2.2.4.2] 要求 [(e)RedCap UE] 使用的 [initial DL/UL BWP] 不得超過 UE 最大頻寬。

[2. Key C Functions / Data Structures]

- `nr_validate_redcap_gnb_frame_parms()`：驗證 [gNB] [NR_DL_FRAME_PARMS]
- `nr_validate_redcap_ue_frame_parms()`：驗證 [UE] [NR_DL_FRAME_PARMS]
- `nr_assert_redcap_fr1_grid_size()`：檢查 [PRB] 是否超出 [20 MHz FR1] 上限
- `nr_init_frame_parms()` / `nr_init_frame_parms_ue()`：初始化入口
- `NR_DL_FRAME_PARMS`：承接 [SCS]、[PRB]、[antenna count]、[freq_range]、[redcap_restricted]

[3. Modification Summary]

- [Modification Point] [nr_parms.c] → [Reason] 將 [RedCap FR1] 規則做成公開 helper → [Before vs. After Comparison] [Before] 不易測試；[After] init path 與 unit test 可共用 → [Discussion Point] 不改 backend 行為，只強化檢查。
- [Modification Point] [nr_phy_init.h] → [Reason] 匯出 helper 宣告 → [Before vs. After Comparison] [Before] 內部 static；[After] 可被 test 呼叫 → [Discussion Point] 不改 production path。
- [Modification Point] [test_nr_frame_params.cpp] → [Reason] 補 RedCap regression tests → [Before vs. After Comparison] [Before] 無 RedCap coverage；[After] 有合法與非法案例 → [Discussion Point] 加 test-only stub 隔離 loader link 依賴。

[4. Test Results Summary Table]

[Test Item]	[Status]	[Code Coverage]	[Modification Logs]
[Focused rebuild: nr_parms.c.o]	[Pass]	[Module compile check]	<code>test_log/build_logs/nr_parms_obj_2026-04-09_13-41-21.log</code>
[Configure tests preset]	[Pass]	[Build tree setup]	<code>test_log/build_logs/cmake_configure_tests_2026-04-09_13-41-21.log</code>

[Test Item]	[Status]	[Code Coverage]	[Modification Logs]
[Build test_nr_frame_params]	[Pass]	[Unit test binary build]	test_log/build_logs/test_nr_frame_params_build_rerun_2026-04-09_13-41-21.log
[Run ctest -R test_nr_frame_params]	[Pass]	[4 new RedCap checks + existing frame-param tests]	test_log/compiler_logs/test_nr_frame_params_2026-04-09_13-41-21.log

[5. 3GPP Specification Mapping]

- [TS 38.306 Clause 4.2.21.1] → [FR1 最大頻寬 20 MHz] → 對應 nr_redcap_fr1_max_prbs() 與 nr_assert_redcap_fr1_grid_size()
- [TS 38.306 Clause 4.2.21.1] → [1Rx mandatory / 2Rx optional / DL layers <= 2 / no UL MIMO] → 對應 nr_validate_redcap_gnb_frame_parms() 與 nr_validate_redcap_ue_frame_parms()
- [TS 38.331 Clause 5.2.2.4.2] → [initial DL/UL BWP 不得超過最大頻寬] → 對應 init-time grid size validation

[6. How To Use This Code]

- [Production Path] 當 [nr-softmodem]、[nr-uesoftmodem]、[nr-ru]、或 [rfsimulator] 進入 init，若偵測到 [RedCap restricted]，就會自動執行這組檢查。
- [Developer Path] 也可以直接呼叫 nr_validate_redcap_gnb_frame_parms() 與 nr_validate_redcap_ue_frame_parms() 做獨立驗證。
- [Focused Rebuild] cmake --preset default，再跑 ninja -C cmake_targets/ran_build/build CMakeFiles/PHY_NR.dir/openair1/PHY/INIT/nr_parms.c.o
- [Focused Unit Test] cmake --preset tests，再跑 cmake --build cmake_targets/ran_build/build_test --target test_nr_frame_params 與 ctest --output-on-failure -R test_nr_frame_params

[7. Practice Exercises]

- [Basic] 為什麼 [RedCap FR1] 的限制要先在 [init-time] 檢查？
- [Applied] 若 [mu = 1] 且 [N_RB_DL = 52]，為什麼它應該被拒絕？
- [Advanced] [half-duplex FDD Type A] 的限制應該放在 [init-time]、[scheduler]、還是 [duplex slot logic]？請說明責任分工。